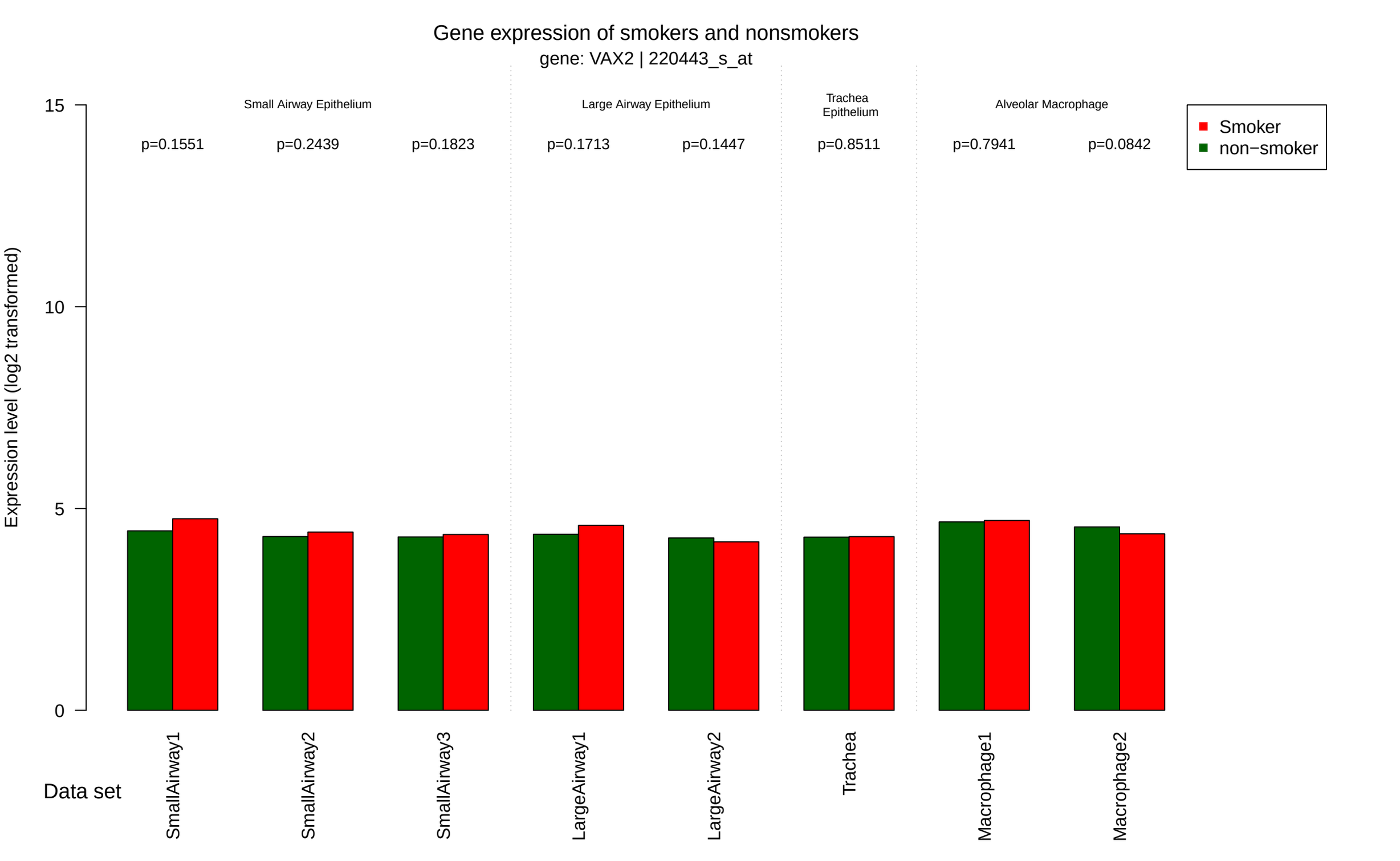
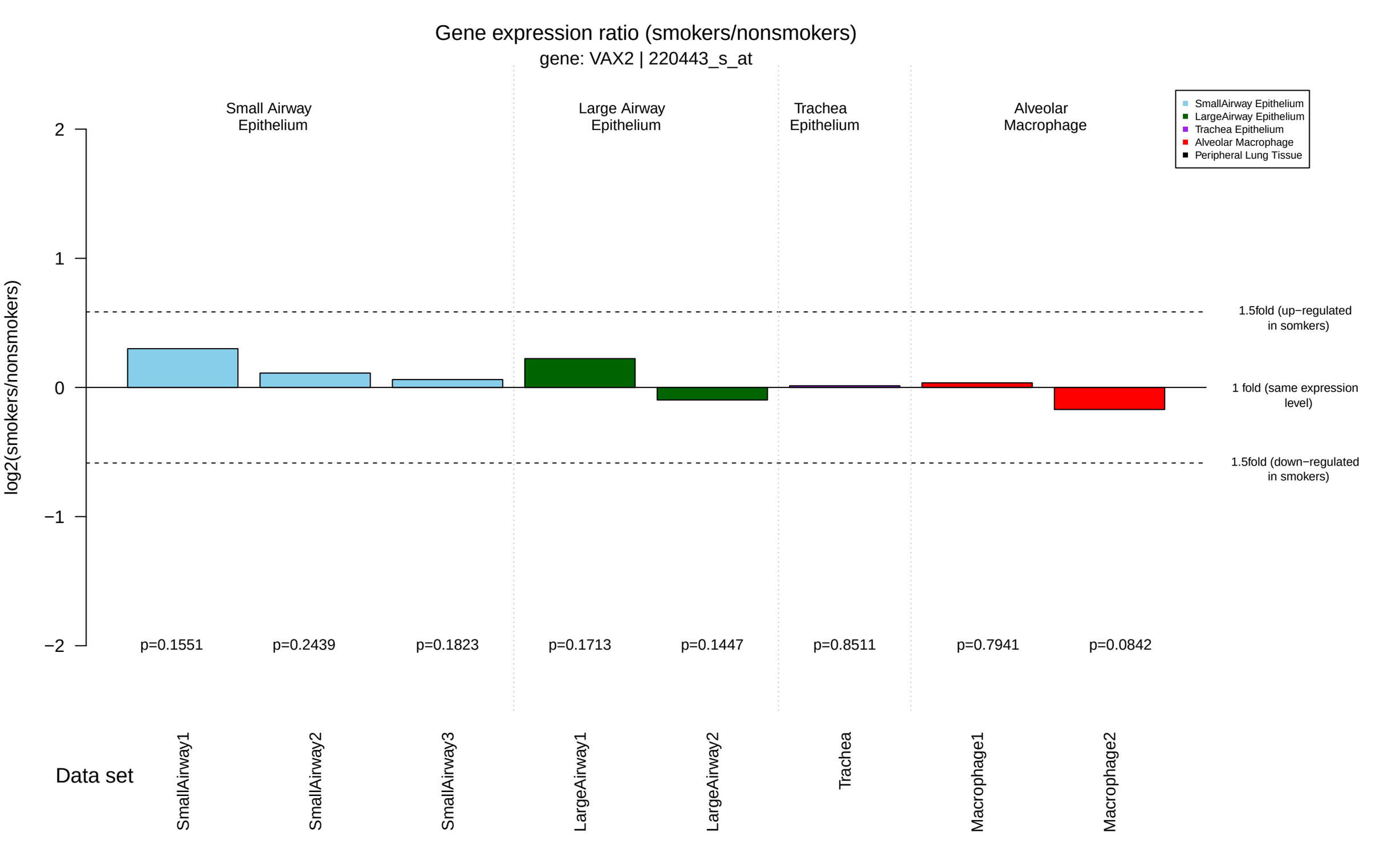
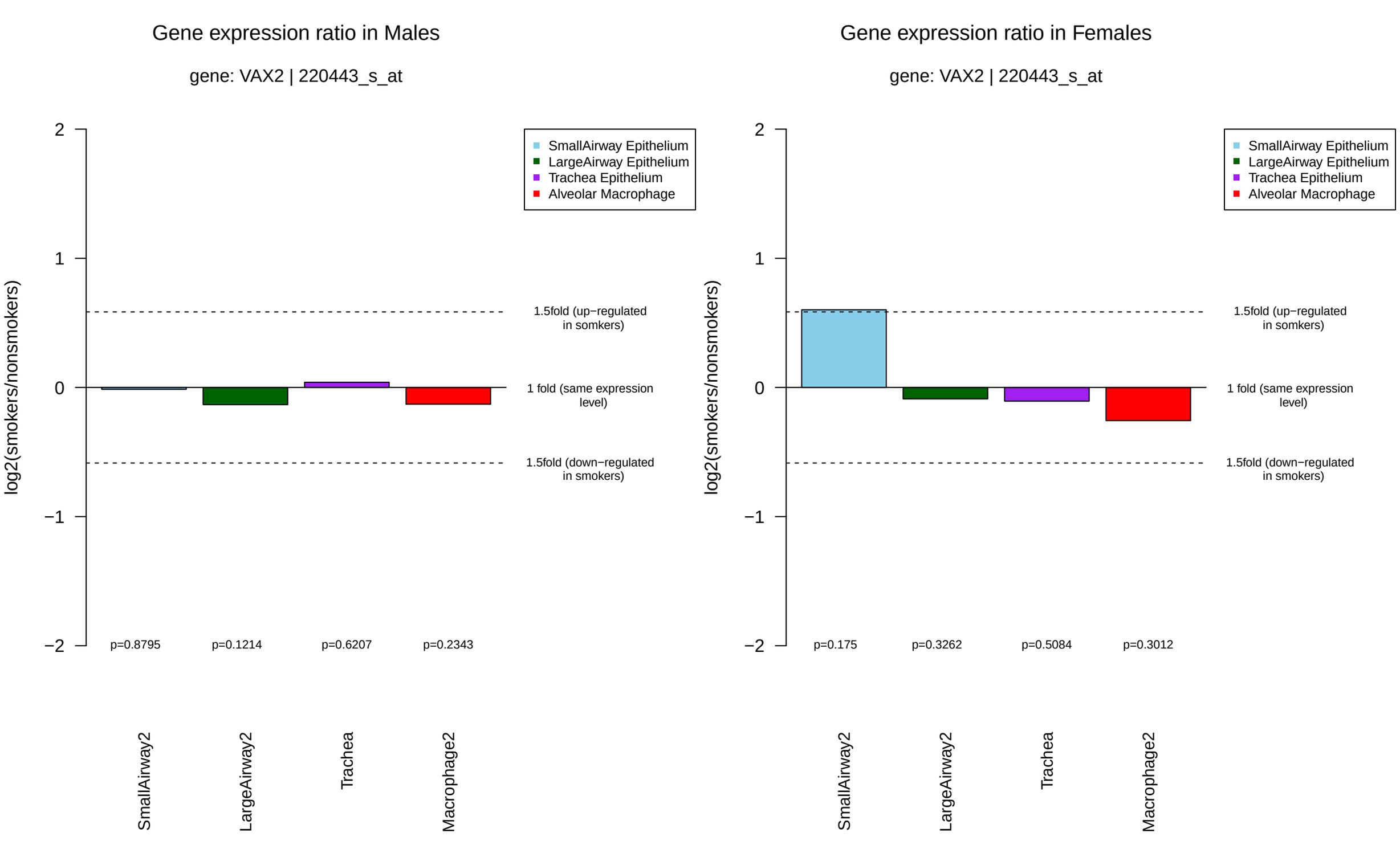
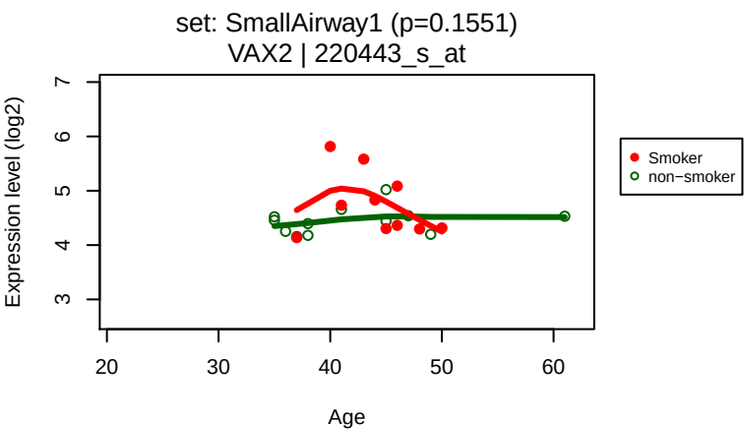
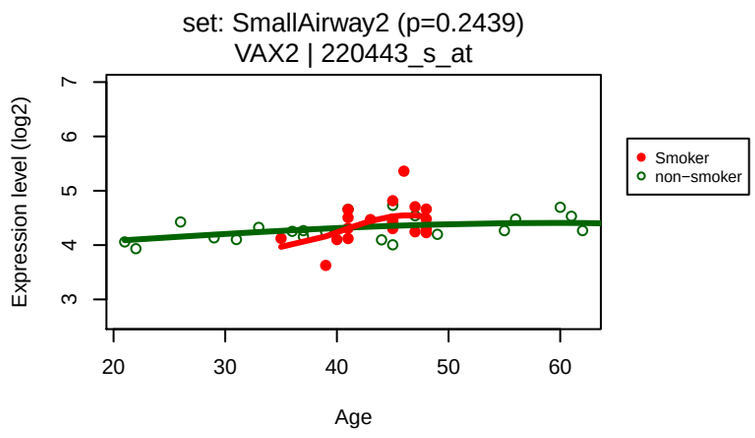
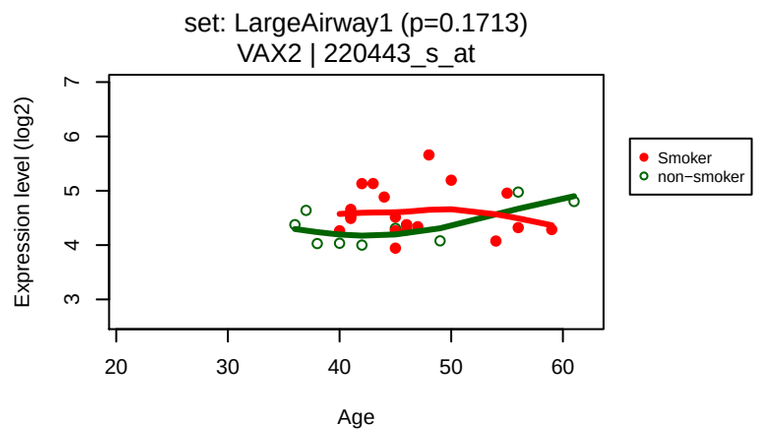

Gene expression of smokers and nonsmokers
gene: VAX2 | 220443_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1551
p=0.2439
p=0.1823
p=0.1713
p=0.1447
p=0.8511
p=0.7941
p=0.0842
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

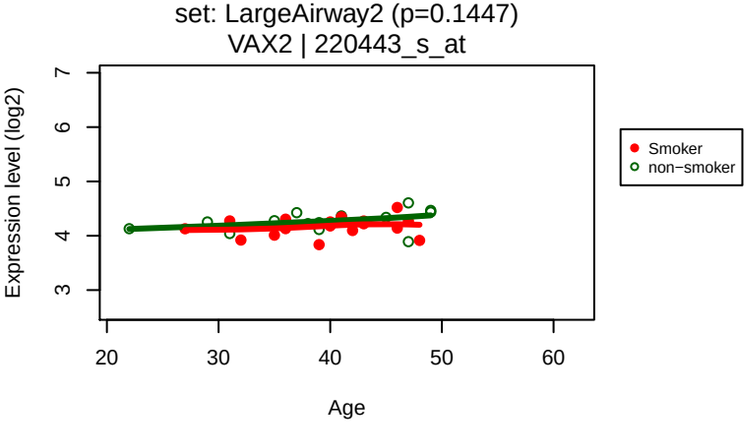

Gene expression ratio (smokers/nonsmokers)
gene: VAX2 | 220443_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1551
p=0.2439
p=0.1823
p=0.1713
p=0.1447
p=0.8511
p=0.7941
p=0.0842
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

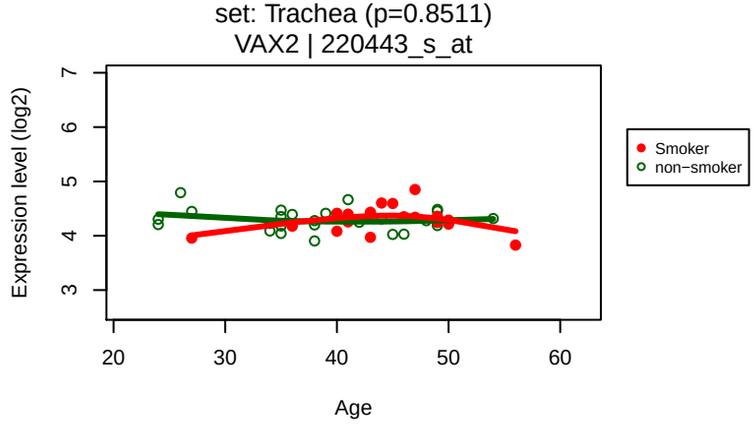

Gene expression ratio in Males
gene: VAX2 | 220443_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8795
p=0.1214
p=0.6207
p=0.2343
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VAX2 | 220443_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.175
p=0.3262
p=0.5084
p=0.3012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

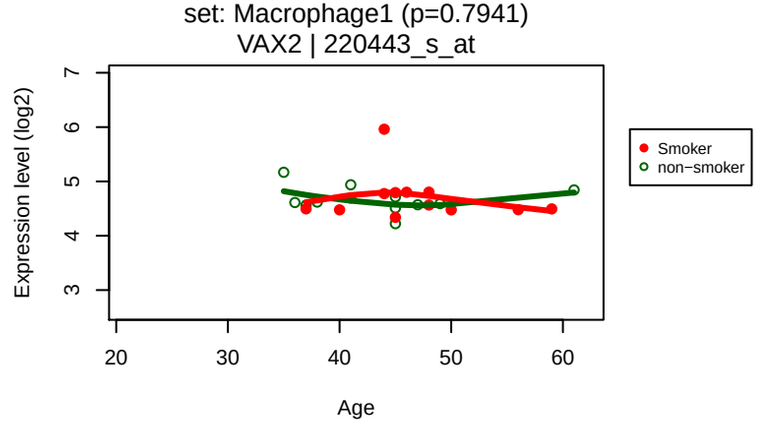

set: SmallAirway1 (p=0.1551)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker

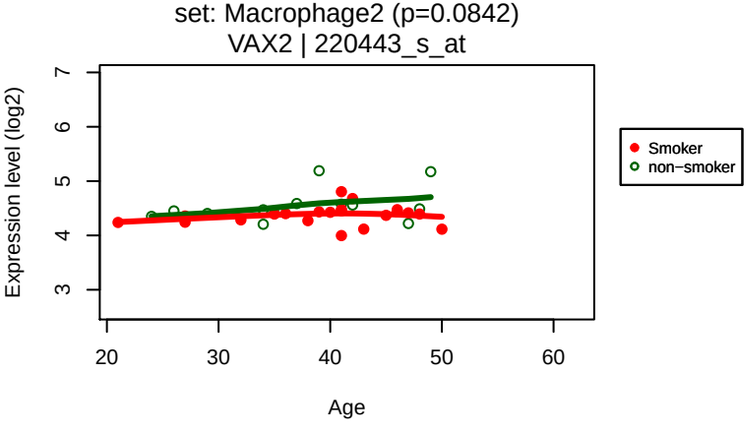

set: SmallAirway2 (p=0.2439)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker

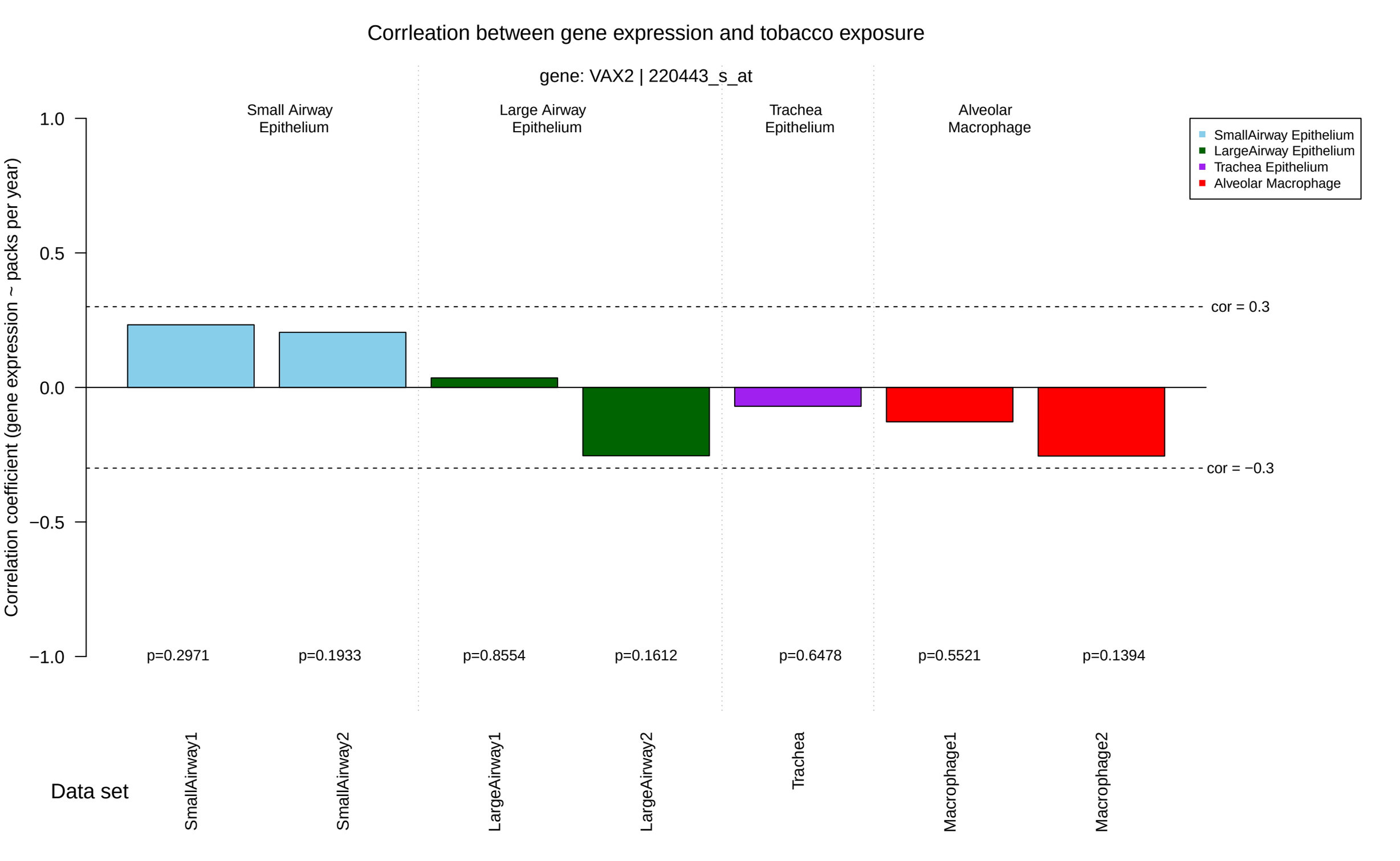

set: LargeAirway1 (p=0.1713)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1447)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8511)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7941)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0842)
VAX2 | 220443_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VAX2 | 220443_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2971
p=0.1933
p=0.8554
p=0.1612
p=0.6478
p=0.5521
p=0.1394
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2